

Quantum-Response and Distortion-Region Simulation for Conventional, LO-Free, and LO-Dressed Rydberg Receivers

QuTiP and Monte-Carlo Implementation Guide for Fig. 9

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1 Purpose of This Document

This document provides the numerical implementation framework required to extend the analytical SER–SNR derivations into the distortion regions of Rydberg atomic receivers.

The previous mathematical document established the analytical operating-region relationships

$$\text{SNR} \longrightarrow \text{SER}. \quad (1)$$

For the conventional receiver this analytical relationship is essentially the complete model.

For the Rydberg receivers, however, the complete communication chain contains an additional quantum-response layer:

$$\boxed{\text{SNR} \longrightarrow P_{\text{Rx}} \longrightarrow E_{\text{RF}} \longrightarrow \Omega_{\text{RF}} \longrightarrow \text{atomic response} \longrightarrow \text{detected symbol} \longrightarrow \text{SER}.} \quad (2)$$

The purpose of this document is not to replace the analytical derivations. Instead, it provides the numerical procedure required when the atomic response becomes nonlinear or the quantum measurement becomes ambiguous.

2 Three Receiver Models

The three receiver configurations considered are:

1. Conventional RF receiver,
2. LO-free Rydberg atomic receiver,
3. LO-dressed Rydberg atomic receiver.

2.1 Conventional Receiver

The conventional receiver is modeled as

$$y = \sqrt{P_{\text{Rx}}}hx + n_{\text{Conv}}, \quad (3)$$

with

$$n_{\text{Conv}} \sim \mathcal{CN}(0, \sigma_{\text{Conv}}^2). \quad (4)$$

For Fig. 9,

$$\boxed{\text{SNR} \longrightarrow \text{SER}_{\text{QAM}}.} \quad (5)$$

No QuTiP simulation is required.

2.2 LO-Free Rydberg Receiver

The LO-free receiver measures RF amplitude through Autler–Townes splitting:

$$\boxed{\text{SNR} \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{EIT spectrum} \rightarrow \text{AT peak separation} \rightarrow \hat{x} \rightarrow \text{SER}.} \quad (6)$$

The weak-RF distortion mechanism occurs when the two AT peaks can no longer be reliably resolved.

The approximately 5.5 MHz value reported by the reference study is treated only as a reference point. The actual simulation threshold is determined numerically.

2.3 LO-Dressed Rydberg Receiver

The LO-dressed receiver uses a strong local oscillator:

$$\boxed{\text{SNR} \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow P_{\text{out}} \rightarrow \text{photodetection} \rightarrow \hat{x} \rightarrow \text{SER}.} \quad (7)$$

Inside the linear region,

$$P_{\text{out}} \approx \bar{P}_0 + \kappa \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi). \quad (8)$$

At sufficiently large RF Rabi frequency, this approximation fails and atomic nonlinear distortion appears.

3 Analytical Region versus Distortion Region

The simulation is divided into an analytical operating region and a numerical distortion region.

3.1 Operating Region

For conventional QAM,

$$\boxed{\text{SER}_{\text{Conv}} = \text{SER}_{\text{QAM}}(\gamma_{\text{Conv}}, M).} \quad (9)$$

For LO-free operation,

$$\boxed{\text{SER}_{\text{LF}} = \text{SER}_{\text{PAM}}(\gamma_{\text{LF}}, M)} \quad (10)$$

provided

$$|\Omega_{\text{RF}}| \geq \Omega_{\text{RF,th}}. \quad (11)$$

For LO-dressed operation,

$$\boxed{\text{SER}_{\text{LO}} = \text{SER}_{\text{QAM}}(\gamma_{\text{LO}}, M)} \quad (12)$$

provided

$$|\Omega_{\text{RF}}| \leq \Omega_{\text{RF,max}}. \quad (13)$$

3.2 Distortion Region

The quantum-response region is obtained numerically:

$$\boxed{\text{Master equation} \rightarrow \text{QuTiP} \rightarrow \text{quantum response} \rightarrow \text{lookup table} \rightarrow \text{Monte-Carlo detector} \rightarrow \text{SER}.} \quad (14)$$

QuTiP is therefore used to characterize the atomic receiver response, not to solve a separate steady-state problem for every Monte-Carlo symbol.

4 Conventional Receiver: Complete Implementation

4.1 Analytical QAM SER

For square M -QAM,

$$\text{SER}_{\text{QAM}} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3\gamma}{M-1}} \right) \right]^2. \quad (15)$$

For an SNR axis in dB,

$$\gamma = 10^{\text{SNR}_{\text{dB}}/10}. \quad (16)$$

4.2 Monte-Carlo Validation

The conventional simulation:

1. generates random QAM symbols,
2. normalizes the constellation to unit average energy,
3. adds complex Gaussian noise,
4. performs nearest-neighbor detection,
5. counts symbol errors.

$$\text{SER}_{\text{MC}} = \frac{N_{\text{errors}}}{N_{\text{symbols}}}.$$

 (17)

4.3 Conventional Python

```
1 import numpy as np
2
3 def square_qam_constellation(M):
4
5     m = int(np.sqrt(M))
6
7     if m * m != M:
8         raise ValueError(
9             "M must be a perfect square."
10        )
11
12    levels = np.arange(
13        -(m - 1),
14        m,
15        2
16    )
17
18    constellation = []
19
20    for i in levels:
21        for q in levels:
22            constellation.append(i + 1j*q)
23
24    constellation = np.asarray(
```

```

25     constellation,
26     dtype=complex
27 )
28
29 # Unit average symbol energy
30 constellation /= np.sqrt(
31     np.mean(np.abs(constellation)**2)
32 )
33
34 return constellation
35
36
37 def qam_detect(y, constellation):
38
39     distances = np.abs(
40         y[:, None]
41         - constellation[None, :]
42     )**2
43
44     return np.argmin(
45         distances,
46         axis=1
47     )
48
49
50 def simulate_conventional_qam(
51     snr_db,
52     M,
53     n_symbols=100000,
54     rng=None
55 ):
56
57     if rng is None:
58         rng = np.random.default_rng()
59
60     constellation = square_qam_constellation(M)
61
62     indices = rng.integers(
63         0,
64         M,
65         n_symbols
66     )
67
68     x = constellation[indices]
69
70     gamma = 10**(
71         snr_db / 10
72     )
73
74     # Unit-energy symbols:
75     #  $E[|n|^2] = 1/\text{gamma}$ 
76     noise_variance = 1.0 / gamma
77
78     noise = np.sqrt(
79         noise_variance / 2
80     ) * (
81         rng.standard_normal(n_symbols)
82         +

```

```

83         1j * rng.standard_normal(n_symbols)
84     )
85
86     y = x + noise
87
88     detected = qam_detect(
89         y,
90         constellation
91     )
92
93     return np.mean(
94         detected != indices
95     )

```

This normalized Monte-Carlo implementation is independent of the physical P_{Rx} model.

5 SNR-to-Received-Power Mapping

This section removes the previously unresolved K_{Ω} scale.

The communication simulation must not introduce an arbitrary relation

$$\Omega_{\text{RF}} = K_{\Omega} x. \quad (18)$$

Instead, each SNR point is first converted to physical received power using the closed-form inversion derived in Document 1.

5.1 LO-Free Inversion

Define

$$\sigma_{\text{LF},0}^2 = \sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2. \quad (19)$$

The LO-free SNR model is

$$\gamma_{\text{LF}} = \frac{P_{\text{Rx}} |h|^2}{\sigma_{\text{LF},0}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}. \quad (20)$$

Therefore,

$$P_{\text{Rx,LF}} = \frac{\gamma_{\text{LF}} \sigma_{\text{LF},0}^2}{|h|^2 - \gamma_{\text{LF}} \tilde{\epsilon}^2}. \quad (21)$$

This inversion is valid only when

$$|h|^2 > \gamma_{\text{LF}} \tilde{\epsilon}^2. \quad (22)$$

Hence the uncertainty-limited SNR ceiling is

$$\gamma_{\text{LF,max}} = \frac{|h|^2}{\tilde{\epsilon}^2}. \quad (23)$$

5.2 LO-Dressed Inversion

Define

$$A_{\text{LO}} = G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}}{\hbar} \right)^2. \quad (24)$$

The linear-region LO-dressed SNR is

$$\gamma_{\text{LO}} = \frac{A_{\text{LO}} P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2}. \quad (25)$$

Therefore,

$$P_{\text{Rx,LO}} = \frac{\gamma_{\text{LO}} \sigma_{\text{Ry,LO}}^2}{A_{\text{LO}} |h|^2}. \quad (26)$$

No numerical root solver is required for this mapping.

5.3 RF Field and Rabi Frequency

After obtaining P_{Rx} ,

$$|E_{\text{RF}}| = \sqrt{P_{\text{Rx}}}. \quad (27)$$

The RF Rabi frequency is

$$|\Omega_{\text{RF}}| = \frac{|\wp_{\text{RF}}|}{\hbar} |E_{\text{RF}}|. \quad (28)$$

Hence

$$|\Omega_{\text{RF}}| = \frac{|\wp_{\text{RF}}|}{\hbar} \sqrt{P_{\text{Rx}}}. \quad (29)$$

The conversion to MHz is only

$$f_{\text{RF}} = \frac{|\Omega_{\text{RF}}|}{2\pi}. \quad (30)$$

5.4 Python Mapping Functions

```

1 def snr_db_to_linear(snr_db):
2     return 10.0**(snr_db / 10.0)
3
4
5 def prx_from_snr_lo_free(
6     snr_db,
7     sigma_bgn_sq,
8     sigma_qpn_sq,
9     h_abs_sq,
10    epsilon_sq
11 ):
12
13     gamma = snr_db_to_linear(snr_db)
14
15     sigma0_sq = (
16         sigma_bgn_sq

```

```

17         + sigma_qpn_sq
18     )
19
20     denominator = (
21         h_abs_sq
22         - gamma * epsilon_sq
23     )
24
25     if denominator <= 0:
26         return np.nan
27
28     return (
29         gamma * sigma0_sq
30         / denominator
31     )
32
33
34 def prx_from_snr_lo_dressed(
35     snr_db,
36     sigma_ry_lo_sq,
37     G_LNA,
38     R_L,
39     D,
40     kappa,
41     mu_RF,
42     hbar,
43     h_abs_sq
44 ):
45
46     gamma = snr_db_to_linear(snr_db)
47
48     A_LO = (
49         G_LNA
50         * R_L
51         * D**2
52         * kappa**2
53         * (mu_RF / hbar)**2
54     )
55
56     return (
57         gamma
58         * sigma_ry_lo_sq
59         / (A_LO * h_abs_sq)
60     )
61
62
63 def prx_to_omega_rf(
64     prx,
65     mu_RF,
66     hbar
67 ):
68
69     if prx < 0:
70         return np.nan
71
72     E_RF = np.sqrt(prx)
73
74     return (

```

```

75     abs(mu_RF)
76     / hbar
77     * E_RF
78 )

```

Thus, there is ****no arbitrary K_Ω **** in the physical simulation.

6 Four-Level Quantum Model

The quantum system contains

$$|1\rangle \rightarrow |2\rangle \rightarrow |3\rangle \rightarrow |4\rangle, \quad (31)$$

corresponding to

$$6S_{1/2} \rightarrow 6P_{3/2} \rightarrow 47D_{5/2} \rightarrow 48P_{3/2}. \quad (32)$$

The probe drives

$$|1\rangle \leftrightarrow |2\rangle,$$

the coupling laser drives

$$|2\rangle \leftrightarrow |3\rangle,$$

and the RF field drives

$$|3\rangle \leftrightarrow |4\rangle.$$

6.1 Hamiltonian

The rotating-frame Hamiltonian is

$$H = \frac{\hbar}{2} \begin{bmatrix} 0 & \Omega_p & 0 & 0 \\ \Omega_p & -2\Delta_p & \Omega_c & 0 \\ 0 & \Omega_c & -2(\Delta_p + \Delta_c) & \Omega \\ 0 & 0 & \Omega & -2(\Delta_p + \Delta_c + \Delta_{\text{RF}}) \end{bmatrix}. \quad (33)$$

QuTiP uses the Hamiltonian in angular-frequency units, so the implementation below omits the common factor $\hbar$.

6.2 Decay Rates

The reference parameters are

$$\gamma_1 = 0, \quad (34)$$

$$\frac{\gamma_2}{2\pi} = 5.2 \text{ MHz}, \quad (35)$$

$$\frac{\gamma_3}{2\pi} = 3.9 \text{ kHz}, \quad (36)$$

$$\frac{\gamma_4}{2\pi} = 1.7 \text{ kHz}. \quad (37)$$

Thus,

$$\gamma_2 = 2\pi(5.2 \times 10^6), \quad (38)$$

$$\gamma_3 = 2\pi(3.9 \times 10^3), \quad (39)$$

$$\gamma_4 = 2\pi(1.7 \times 10^3). \quad (40)$$

6.3 Dipole Moments

The RF transition dipole is

$$\wp_{\text{RF}} = -1443.459 \, ea_0. \quad (41)$$

The probe transition dipole is

$$\boxed{\wp_{12} = (2.5 \, ea_0)^2}. \quad (42)$$

The square is intentional and must be retained exactly as required by the parameterization used in the analytical model.

6.4 Laser Parameters

The reference configuration uses

$$\frac{\Omega_p}{2\pi} = 8 \text{ MHz}, \quad (43)$$

and

$$\frac{\Omega_c}{2\pi} = 1 \text{ MHz}. \quad (44)$$

7 QuTiP Implementation Architecture

The implementation consists of

1. physical constants,
2. quantum parameters,
3. basis states,
4. Hamiltonian,
5. collapse operators,
6. steady-state solution,
7. optical-response extraction,
8. parameter sweep,
9. lookup-table generation,
10. Monte-Carlo communication simulation.

The central computational rule is

$$\boxed{\text{QuTiP characterization} \rightarrow \text{lookup table} \rightarrow \text{Monte Carlo}}. \quad (45)$$

7.1 Imports

```
1 import numpy as np
2 import qutip as qt
3 import matplotlib.pyplot as plt
4
5 from scipy.signal import find_peaks
6 from scipy.interpolate import interp1d
7 from scipy.stats import linregress
```

7.2 Quantum Parameters

```
1 import numpy as np
2
3 hbar = 1.054571817e-34
4
5 e_charge = 1.602176634e-19
6 a0 = 5.29177210903e-11
7
8 gamma1 = 0.0
9 gamma2 = 2*np.pi*5.2e6
10 gamma3 = 2*np.pi*3.9e3
11 gamma4 = 2*np.pi*1.7e3
12
13 Omega_p = 2*np.pi*8e6
14 Omega_c = 2*np.pi*1e6
15
16 mu_RF = -1443.459 * e_charge * a0
17
18 # IMPORTANT:
19 # The parameter is intentionally squared.
20 mu_12 = (2.5 * e_charge * a0)**2
```

8 Constructing the Four-Level Hamiltonian

```
1 def basis_states():
2
3     ket1 = qt.basis(4, 0)
4     ket2 = qt.basis(4, 1)
5     ket3 = qt.basis(4, 2)
6     ket4 = qt.basis(4, 3)
7
8     return ket1, ket2, ket3, ket4
9
10
11 def four_level_hamiltonian(
12     Omega_p,
13     Omega_c,
14     Omega_RF,
15     Delta_p,
16     Delta_c,
17     Delta_RF
18 ):
19
20     ket1, ket2, ket3, ket4 = basis_states()
```

```

21
22 P2 = ket2 * ket2.dag()
23 P3 = ket3 * ket3.dag()
24 P4 = ket4 * ket4.dag()
25
26 H = (
27     -Delta_p * P2
28     -(Delta_p + Delta_c) * P3
29     -(Delta_p + Delta_c + Delta_RF) * P4
30 )
31
32 H += (
33     Omega_p / 2
34     * (
35         ket1 * ket2.dag()
36         +
37         ket2 * ket1.dag()
38     )
39 )
40
41 H += (
42     Omega_c / 2
43     * (
44         ket2 * ket3.dag()
45         +
46         ket3 * ket2.dag()
47     )
48 )
49
50 H += (
51     Omega_RF / 2
52     * (
53         ket3 * ket4.dag()
54         +
55         ket4 * ket3.dag()
56     )
57 )
58
59 return H

```

9 Decay Operators

For computational implementation we use the simplified cascade

$$|2\rangle \rightarrow |1\rangle, \quad |3\rangle \rightarrow |2\rangle, \quad |4\rangle \rightarrow |3\rangle. \quad (46)$$

The collapse operators are

$$C_2 = \sqrt{\gamma_2}|1\rangle\langle 2|, \quad (47)$$

$$C_3 = \sqrt{\gamma_3}|2\rangle\langle 3|, \quad (48)$$

$$C_4 = \sqrt{\gamma_4}|3\rangle\langle 4|. \quad (49)$$

```

1 def collapse_operators():
2

```

```

3 ket1, ket2, ket3, ket4 = basis_states()
4
5 c2 = np.sqrt(gamma2) * ket1 * ket2.dag()
6 c3 = np.sqrt(gamma3) * ket2 * ket3.dag()
7 c4 = np.sqrt(gamma4) * ket3 * ket4.dag()
8
9 return [c2, c3, c4]

```

Implementation disclosure: this is a simplified cascade Lindblad model. It is not claimed to be the exact full Lindblad superoperator containing every cross-coupling coherence-decay term of the reference formulation. This approximation must therefore be stated explicitly when reporting numerical results.

10 Steady-State Quantum Solution

$$\rho_{ss} = \text{steadystate}(H, \{C_i\}). \quad (50)$$

```

1 def steady_state_rho(
2     Omega_p,
3     Omega_c,
4     Omega_RF,
5     Delta_p,
6     Delta_c,
7     Delta_RF
8 ):
9
10    H = four_level_hamiltonian(
11        Omega_p,
12        Omega_c,
13        Omega_RF,
14        Delta_p,
15        Delta_c,
16        Delta_RF
17    )
18
19    rho_ss = qt.steadystate(
20        H,
21        collapse_operators()
22    )
23
24    return rho_ss

```

11 Optical Response

The required density-matrix element is

$$\rho_{21} = \langle 2|\rho|1\rangle. \quad (51)$$

In QuTiP:

```

1 def rho21_from_state(rho):
2     return rho[1, 0]

```

The exact transmission normalization must be kept consistent with the analytical model.

12 PART A – LO-Free Distortion Region

12.1 Physical Mechanism

The LO-free receiver determines RF amplitude from AT splitting:

$$\Delta\nu_{\text{AT}} \approx \frac{|\Omega_{\text{RF}}|}{2\pi} \quad (52)$$

in the resolvable strong-field region.

At weak RF fields, the two peaks become unresolved.

12.2 EIT Spectrum

```
1 def coupling_detuning_grid():
2
3     return 2*np.pi*np.linspace(
4         -10e6,
5         10e6,
6         2001
7     )
8
9
10 def lo_free_eit_spectrum(
11     Omega_RF,
12     delta_p=0.0
13 ):
14
15     delta_c_values = (
16         coupling_detuning_grid()
17     )
18
19     spectrum = []
20
21     for delta_c in delta_c_values:
22
23         rho_ss = steady_state_rho(
24             Omega_p,
25             Omega_c,
26             Omega_RF,
27             delta_p,
28             delta_c,
29             0.0
30         )
31
32         rho21 = rho21_from_state(
33             rho_ss
34         )
35
36         spectrum.append(
37             np.imag(rho21)
38         )
39
40     return (
41         delta_c_values,
42         np.asarray(spectrum)
43     )
```


12.3 Sign-Convention Validation

The sign of $\text{Im}(\rho_{21})$ depends on the convention used for the density matrix and dipole interaction. Therefore, before using an automatic peak finder, explicitly plot the spectrum at

$$\frac{\Omega_{\text{RF}}}{2\pi} = 6 \text{ MHz.} \quad (53)$$

The implementation must visually verify that the selected quantity produces the two expected AT features.

Only after this validation should ‘*find_{peaks}*’ be used.

12.4 AT Peak Detection

```
1 def find_at_splitting(  
2     delta_c,  
3     spectrum,  
4     use_absorption=True  
5 ):  
6  
7     response = np.asarray(  
8         spectrum  
9     )  
10  
11     if not use_absorption:  
12         response = -response  
13  
14     peaks, properties = find_peaks(  
15         response,  
16         prominence=0.01  
17     )  
18  
19     if len(peaks) < 2:  
20         return np.nan  
21  
22     strongest = peaks[  
23         np.argsort(  
24             response[peaks]  
25         )[-2:]]  
26  
27  
28     strongest = np.sort(  
29         strongest  
30     )  
31  
32     return abs(  
33         delta_c[strongest[1]]  
34         -  
35         delta_c[strongest[0]]  
36     )
```

The sign convention must be fixed from the validation spectrum before the final lookup table is generated.

12.5 LO-Free Characterization

```
1 def characterize_lo_free(  
    ...
```

```

2     omega_rf_values ,
3     use_absorption=True
4 ):
5
6     measured_splitting = []
7
8     for Omega_RF in omega_rf_values:
9
10        delta_c, spectrum = (
11            lo_free_eit_spectrum(
12                Omega_RF
13            )
14        )
15
16        splitting = (
17            find_at_splitting(
18                delta_c,
19                spectrum,
20                use_absorption
21            )
22        )
23
24        measured_splitting.append(
25            splitting
26        )
27
28    return np.asarray(
29        measured_splitting
30    )

```

Example grid:

```

1 omega_rf_values = 2*np.pi*np.linspace(
2     1e6,
3     10e6,
4     181
5 )

```

Save the lookup table:

```

1 np.savez(
2     "lo_free_quantum_response.npz",
3     omega_rf=omega_rf_values,
4     measured_splitting=measured_splitting
5 )

```

12.6 LO-Free Threshold

Define

$$\epsilon_{\text{AT}} = \frac{|\Delta\nu_{\text{QuTiP}} - \Delta\nu_{\text{theory}}|}{\Delta\nu_{\text{theory}}}. \quad (54)$$

A documented tolerance

$$\epsilon_{\text{AT}} \leq \epsilon_{\text{tol}} \quad (55)$$

defines the operating region.

Thus,

$$\Omega_{\text{RF,th}} = \text{lowest RF Rabi frequency satisfying the chosen AT-resolution criterion.} \quad (56)$$

The approximately 5.5 MHz value must therefore be obtained from the simulation rather than simply inserted into the final SER calculation.

12.7 LO-Free Lookup Table

```

1 def build_lo_free_interpolator(
2     omega_rf,
3     measured_splitting
4 ):
5
6     valid = np.isfinite(
7         measured_splitting
8     )
9
10    return interp1d(
11        omega_rf[valid],
12        measured_splitting[valid],
13        kind="linear",
14        bounds_error=True
15    )

```

No extrapolation should be used outside the QuTiP-generated range.

13 LO-Free Distortion-to-SER Simulation

The symbol-level chain is

$$x_k \rightarrow P_{\text{Rx},k} \rightarrow E_{\text{RF},k} \rightarrow \Omega_{\text{RF},k} \rightarrow \text{AT response} \rightarrow \widehat{\Omega}_{\text{RF},k} \rightarrow \hat{x}_k. \quad (57)$$

The physical RF mapping is obtained from the SNR inversion of Document 1. There is therefore no independent K_{Ω} parameter.

13.1 PAM Levels

```

1 def pam_levels(M):
2
3     levels = np.arange(
4         -(M - 1),
5         M,
6         2,
7         dtype=float
8     )
9
10    levels /= np.sqrt(
11        np.mean(levels**2)
12    )
13
14    return levels

```

13.2 SNR-to-RF-Rabi Mapping

For each SNR point:

```
1 def lo_free_omega_from_snr(  
2     snr_db,  
3     sigma_bgn_sq,  
4     sigma_qpn_sq,  
5     h_abs_sq,  
6     epsilon_sq,  
7     mu_RF,  
8     hbar  
9 ):  
10  
11     prx = prx_from_snr_lo_free(  
12         snr_db,  
13         sigma_bgn_sq,  
14         sigma_qpn_sq,  
15         h_abs_sq,  
16         epsilon_sq  
17     )  
18  
19     if not np.isfinite(prx):  
20         return np.nan  
21  
22     return prx_to_omega_rf(  
23         prx,  
24         mu_RF,  
25         hbar  
26     )
```

For a normalized PAM symbol x_k , the physical amplitude mapping must be applied consistently to the chosen received-power reference.

13.3 LO-Free Quantum Detector

```
1 def lo_free_quantum_detect(  
2     omega_rf_symbols,  
3     pam_levels_abs,  
4     at_interpolator,  
5     splitting_noise_std,  
6     rng  
7 ):  
8  
9     estimated_splitting = (  
10         at_interpolator(  
11             omega_rf_symbols  
12         )  
13     )  
14  
15     estimated_splitting += (  
16         splitting_noise_std  
17         * rng.standard_normal(  
18             len(omega_rf_symbols)  
19         )  
20     )  
21  
22     estimated_omega = (  
23         estimated_splitting
```

```

23     2*np.pi
24     * estimated_splitting
25 )
26
27     estimated_amplitude = np.abs(
28         estimated_omega
29     )
30
31     distances = np.abs(
32         estimated_amplitude[:, None]
33         -
34         pam_levels_abs[None, :]
35     )
36
37     return np.argmin(
38         distances,
39         axis=1
40     )

```

The physical noise calibration must be obtained from the LO-free noise model rather than from an arbitrary normalized noise value.

14 LO-Free Final Piecewise Model

$$\text{SER}_{\text{LF}} = \begin{cases} \text{SER}_{\text{PAM}}(\gamma_{\text{LF}}, M), & |\Omega_{\text{RF}}| \geq \Omega_{\text{RF,th}}, \\ \text{SER}_{\text{MC,QuTiP}}(\Omega_{\text{RF}}, M), & |\Omega_{\text{RF}}| < \Omega_{\text{RF,th}}. \end{cases} \quad (58)$$

15 PART B – LO-Dressed Quantum Response

15.1 Physical Principle

The LO-dressed receiver uses a strong LO and a weak RF signal.

The small-signal response is

$$P_{\text{out}}(t) \approx \bar{P}_0 + \kappa \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi). \quad (59)$$

The numerical objective is to determine where this approximation ceases to be sufficiently linear.

15.2 Reference Operating Point

The reference configuration uses approximately

$$\frac{\Omega_{\text{LO}}}{2\pi} = 4.23 \text{ MHz}, \quad (60)$$

with

$$\frac{\Omega_{\text{RF}}}{2\pi} = 20 \text{ kHz} \quad (61)$$

and

$$\Delta f = 150 \text{ kHz}. \quad (62)$$

The LO operating point is therefore explicitly retained in the linear-region search.

15.3 Total Rabi Frequency

For the LO-dressed characterization,

$$\Omega_{\text{total}} = \Omega_{\text{LO}} + \Omega_{\text{RF}} \quad (63)$$

for the in-phase scalar characterization convention.

For a full complex-envelope treatment, the phase of the RF field must also be retained.

15.4 LO-Dressed Sweep

```
1 omega_lo = 2*np.pi*4.23e6
2
3 omega_total_values = (
4     2*np.pi
5     * np.linspace(
6         0.0,
7         10e6,
8         401
9     )
10 )
```

The sweep must contain the LO operating point.

15.5 Probe Response

```
1 def probe_response_from_rho21(
2     rho21,
3     alpha=1.0
4 ):
5
6     absorption = np.imag(
7         rho21
8     )
9
10    transmission = np.exp(
11        -alpha * absorption
12    )
13
14    return transmission
```

The exact optical normalization must be calibrated against the reference model.

16 LO-Dressed Linear Dynamic Range

The critical correction is that the linear fit must be centered on the physical LO operating point.

The search must therefore be performed around

$$\Omega_{\text{total}} \approx \Omega_{\text{LO}} = 2\pi(4.23 \text{ MHz}). \quad (64)$$

It is incorrect to determine the receiver's useful linear region by simply fitting from $\Omega_{\text{total}} = 0$.

16.1 Linear-Fit Criterion

For a candidate interval,

$$P_{\text{out}} \approx a + b\Omega_{\text{total}}. \quad (65)$$

The coefficient of determination is

$$R^2 = 1 - \frac{\sum_i (P_i - \hat{P}_i)^2}{\sum_i (P_i - \bar{P})^2}. \quad (66)$$

```
1 def linear_fit_score(  
2     omega_values,  
3     response_values  
4 ):  
5  
6     slope, intercept, r_value, _, _ = (  
7         linregress(  
8             omega_values,  
9             response_values  
10        )  
11    )  
12  
13    predicted = (  
14        intercept  
15        +  
16        slope * omega_values  
17    )  
18  
19    residual = (  
20        response_values  
21        -  
22        predicted  
23    )  
24  
25    rmse = np.sqrt(  
26        np.mean(residual**2)  
27    )  
28  
29    return {  
30        "slope": slope,  
31        "intercept": intercept,  
32        "r2": r_value**2,  
33        "rmse": rmse  
34    }
```

16.2 Operating-Point-Anchored Search

```
1 def find_linear_dynamic_range(  
2     omega_values,  
3     response_values,  
4     omega_lo,  
5     r2_threshold=0.995  
6 ):  
7  
8     omega_values = np.asarray(  
9         omega_values
```

```

10     )
11
12     response_values = np.asarray(
13         response_values
14     )
15
16     center = np.argmin(
17         np.abs(
18             omega_values
19             - omega_lo
20         )
21     )
22
23     best_left = center
24     best_right = center
25     best_fit = None
26
27     for radius in range(
28         5,
29         min(center,
30             len(omega_values)-center-1)
31         + 1
32     ):
33
34         left = center - radius
35         right = center + radius + 1
36
37         omega_test = (
38             omega_values[left:right]
39         )
40
41         response_test = (
42             response_values[left:right]
43         )
44
45         fit = linear_fit_score(
46             omega_test,
47             response_test
48         )
49
50         if fit["r2"] >= r2_threshold:
51
52             best_left = left
53             best_right = right
54             best_fit = fit
55
56         else:
57             break
58
59     if best_fit is None:
60         raise RuntimeError(
61             "No acceptable linear region found "
62             "around the LO operating point."
63         )
64
65     return (
66         omega_values[best_left],
67         omega_values[best_right - 1],

```



```

68     best_fit
69 )

```

The resulting upper RF excursion relative to the LO operating point is

$$\Omega_{\text{RF,max}} = \Omega_{\text{total,max}} - \Omega_{\text{LO}}. \quad (67)$$

This is the quantity used to define the LO-dressed operating region.

17 LO-Dressed Nonlinear Distortion Simulation

The communication chain is

$$x_k \rightarrow \Omega_{\text{RF}}, k \rightarrow P_{\text{out},k} \rightarrow \text{detector} \rightarrow \hat{x}_k. \quad (68)$$

The QuTiP nonlinear response is generated before Monte Carlo.

17.1 LO-Dressed Lookup Table

Generate

$$\{\Omega_{\text{RF},i}, P_{\text{out},i}\}. \quad (69)$$

Then interpolate only inside the simulated range.

```

1 def build_lo_dressed_interpolator(
2     omega_rf,
3     pout
4 ):
5
6     return interp1d(
7         omega_rf,
8         pout,
9         kind="cubic",
10        bounds_error=True
11    )

```

17.2 SNR-to-RF-Rabi Mapping

The LO-dressed communication simulation uses

$$\gamma \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}}. \quad (70)$$

```

1 def lo_dressed_omega_from_snr(
2     snr_db,
3     sigma_ry_lo_sq,
4     G_LNA,
5     R_L,
6     D,
7     kappa,
8     mu_RF,
9     hbar,
10    h_abs_sq
11 ):
12
13    prx = prx_from_snr_lo_dressed(

```

```

14     snr_db,
15     sigma_ry_lo_sq,
16     G_LNA,
17     R_L,
18     D,
19     kappa,
20     mu_RF,
21     hbar,
22     h_abs_sq
23 )
24
25 return prx_to_omega_rf(
26     prx,
27     mu_RF,
28     hbar
29 )

```

Thus the previously unresolved arbitrary scale factor K_Ω is eliminated.

17.3 QAM Symbols

For

$$x_k = I_k + jQ_k, \quad (71)$$

the amplitude is

$$|x_k| \quad (72)$$

and phase is

$$\phi_k = \arg(x_k). \quad (73)$$

The RF Rabi frequency is obtained from the physical received field, not from an arbitrary symbol-to-Rabi constant.

17.4 Nonlinear Detection

The nonlinear atomic response can be represented as

$$P_{\text{out},k} = F_{\text{QuTiP}}(\Omega_{\text{LO}}, \Omega_{\text{RF}}, k, \phi_k). \quad (74)$$

The detector applies the inverse calibrated response:

$$\hat{x}_k = F^{-1}[P_{\text{out},k} + n_k]. \quad (75)$$

In the linear region,

$$F^{-1}(P_{\text{out}}) \approx \frac{P_{\text{out}} - \bar{P}_0}{\kappa}. \quad (76)$$

Outside the linear region, the QuTiP-derived calibration must be used.

18 LO-Dressed Noise Model

The LO-dressed noise contains background, photon shot noise, and thermal noise:

$$\sigma_{\text{Ry,LO}}^2 = \sigma_{\text{BGN,LO}}^2 + \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2. \quad (77)$$

Photon shot noise is

$$\sigma_{\text{PSN}}^2 = 2eB\bar{I}. \quad (78)$$

Thermal noise is

$$\sigma_{\text{TN}}^2 = 4k_B T B. \quad (79)$$

The mean photocurrent must be tied to the same average optical power $\bar{P}_0$ used in Document 1:

$$\bar{I} = \frac{\eta_{\text{eff}} \bar{P}_0}{\hbar \omega_p} e. \quad (80)$$

The LO-dressed linear SNR is

$$\gamma_{\text{LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\omega_{\text{RF}}}{\hbar} \right)^2 P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2}. \quad (81)$$

19 LO-Dressed Final Piecewise Model

$$\text{SER}_{\text{LO}} = \begin{cases} \text{SER}_{\text{QAM}}(\gamma_{\text{LO}}, M), & |\Omega_{\text{RF}}| \leq \Omega_{\text{RF,max}}, \\ \text{SER}_{\text{MC,QuTiP}}(\Omega_{\text{RF}}, M), & |\Omega_{\text{RF}}| > \Omega_{\text{RF,max}}. \end{cases} \quad (82)$$

The physical mechanism is

$$\text{SNR} \uparrow \Rightarrow P_{\text{Rx}} \uparrow \Rightarrow E_{\text{RF}} \uparrow \Rightarrow \Omega_{\text{RF}} \uparrow \Rightarrow \text{nonlinear atomic response} \Rightarrow \text{SER} \uparrow. \quad (83)$$

20 Important Difference Between LO-Free and LO-Dressed

Table 1: Quantum-receiver distortion mechanisms

Receiver	Limitation	QuTiP quantity
Conventional	None in present model	Not required
LO-free	Unresolved AT splitting	AT peak separation
LO-dressed	Nonlinear atomic response	Probe transmission

Therefore,

$$\text{LO-free distortion} = \text{AT measurement ambiguity}. \quad (84)$$

whereas

$$\text{LO-dressed distortion} = \text{atomic nonlinearity}. \quad (85)$$

21 Complete Simulation Architecture

```
rydberg_ser/
|
|-- main.py
|-- config.py
|
|-- quantum/
|   |-- four_level.py
|   |-- lo_free.py
|   |-- lo_dressed.py
|   |-- peak_detection.py
|   |-- response_tables.py
|
|-- communication/
|   |-- qam.py
|   |-- pam.py
|   |-- noise.py
|   |-- conventional.py
|   |-- lo_free_mc.py
|   |-- lo_dressed_mc.py
|
|-- results/
|   |-- lo_free_quantum_response.npz
|   |-- lo_dressed_quantum_response.npz
|   |-- lo_free_ser.npz
|   |-- lo_dressed_ser.npz
|
|-- plots/
|   |-- lo_free_at_splitting.png
|   |-- lo_dressed_dynamic_range.png
|   |-- ser_fig9.png
|
|-- validation/
|   |-- validate_conventional.py
|   |-- validate_lo_free.py
|   |-- validate_lo_dressed.py
|
'-- requirements.txt
```

22 Recommended Execution Order

Stage 1: Conventional Validation

Verify

$$\text{SER}_{\text{MC}} \approx \text{SER}_{\text{Proakis}}. \quad (86)$$

No QuTiP is involved.

Stage 2: Four-Level QuTiP Test

Verify

$$\text{Tr}(\rho) = 1 \quad (87)$$

and

$$\rho = \rho^\dagger. \quad (88)$$

Stage 3: LO-Free EIT Spectrum

For fixed Ω_{RF} , scan Δ_c and reproduce the expected AT structure.

Stage 4: LO-Free Characterization

Sweep Ω_{RF} and calculate

$$\boxed{\Omega_{\text{RF}}/(2\pi) \text{ versus } \Delta\nu_{\text{AT}}.} \quad (89)$$

Stage 5: LO-Free Threshold

Determine $\Omega_{\text{RF,th}}$ using the documented AT-resolution criterion.

Stage 6: LO-Free Monte Carlo

For every SNR point:

$$\text{SNR} \rightarrow \gamma \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{QuTiP lookup table.} \quad (90)$$

Stage 7: LO-Dressed Response

Fix Ω_{LO} and sweep the total RF/Rydberg coupling.

Stage 8: LO-Dressed Linear Range

Determine the linear region around

$$\Omega_{\text{LO}}/(2\pi) = 4.23 \text{ MHz.} \quad (91)$$

Do not fit from $\Omega_{\text{total}} = 0$.

Stage 9: LO-Dressed Monte Carlo

Use the physical

$$\text{SNR} \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \quad (92)$$

mapping and then the QuTiP nonlinear lookup table.

Stage 10: Final Fig. 9

Combine the analytical operating region with the numerical distortion region.

23 Final Fig. 9 Mathematical Structure

The conventional curve is

$$\boxed{\text{SER}_{\text{Conv}} = \text{SER}_{\text{QAM}}(\gamma_{\text{Conv}}, M).} \quad (93)$$

The LO-free curve is

$$\boxed{\text{SER}_{\text{LF}} = \begin{cases} \text{SER}_{\text{PAM}}(\gamma_{\text{LF}}, M), & |\Omega_{\text{RF}}| \geq \Omega_{\text{RF,th}}, \\ \text{SER}_{\text{MC,QuTiP}}, & |\Omega_{\text{RF}}| < \Omega_{\text{RF,th}}. \end{cases}} \quad (94)$$

The LO-dressed curve is

$$\boxed{\text{SER}_{\text{LO}} = \begin{cases} \text{SER}_{\text{QAM}}(\gamma_{\text{LO}}, M), & |\Omega_{\text{RF}}| \leq \Omega_{\text{RF,max}}, \\ \text{SER}_{\text{MC,QuTiP}}, & |\Omega_{\text{RF}}| > \Omega_{\text{RF,max}}. \end{cases}} \quad (95)$$

24 Important Simulation Rules

1. Do not replace the QuTiP response with an arbitrary polynomial.
2. Do not manually draw the distortion region.
3. Do not hard-code the 5.5 MHz LO-free threshold without validating the QuTiP response.
4. Do not use the linear LO-dressed model outside the measured linear dynamic range.
5. Do not call QuTiP once per Monte-Carlo symbol.
6. Generate quantum-response lookup tables first.
7. Use interpolation only inside the simulated response range.
8. Do not introduce an arbitrary K_{Ω} .
9. Use the closed-form SNR-to- P_{Rx} inversion first.
10. Convert P_{Rx} to E_{RF} and then to Ω_{RF} .
11. Keep all Rabi frequencies in rad/s internally.
12. Only convert to MHz for plotting.
13. Keep physical receiver SNR distinct from the E_s/N_0 quantity used by analytical Proakis equations.
14. Use identical constellation normalization in analytical and Monte-Carlo SER calculations.
15. Use sufficiently large Monte-Carlo sample sizes for small SER values.
16. Document every numerical criterion used to define a distortion boundary.

25 Validation Checklist

Conventional

- QAM constellation generated correctly.
- Average symbol energy normalized consistently.
- AWGN variance matches SNR definition.
- Monte-Carlo SER agrees with Proakis.

LO-Free

- Four-level Hamiltonian verified.
- Decay rates verified.
- $\Omega_p/(2\pi) = 8$ MHz verified.
- $\Omega_c/(2\pi) = 1$ MHz verified.
- RF dipole moment verified.
- $\wp_{12} = (2.5ea_0)^2$ retained.
- EIT spectrum obtained.
- Sign convention of $\text{Im}(\rho_{21})$ validated.
- AT peaks detected.
- AT splitting reproduced.
- Threshold extracted numerically.
- QuTiP lookup table saved.
- Physical SNR-to- P_{Rx} mapping used.
- Monte-Carlo distortion simulation implemented.

LO-Dressed

- LO Rabi frequency verified.
- RF Rabi frequency verified.
- Δf verified.
- $\wp_{12}$ square retained.
- Probe transmission calculated.
- LO operating point included in the sweep.
- Linear response region identified around Ω_{LO} .
- Nonlinear response observed.
- $\Omega_{\text{RF,max}}$ extracted.

- Photon shot noise tied to $\bar{P}_0$.
- Physical SNR-to- P_{Rx} mapping used.
- QuTiP lookup table saved.
- QAM Monte-Carlo simulation implemented.

26 Important Limitation of the Reconstruction

The QuTiP procedure reproduces the methodology of the reference study.

An exact pixel-for-pixel reproduction of Fig. 9 should not be claimed until all relevant parameters are matched, including:

1. quantum-system parameters,
2. density-matrix convention,
3. optical-response normalization,
4. peak-detection rule,
5. AT-resolution criterion,
6. LO-dressed linearity criterion,
7. receiver noise calibration,
8. SNR definition,
9. modulation normalization,
10. Monte-Carlo sample size,
11. exact parameters used to generate the original figure.

In particular, the LO-dressed linear dynamic range is a numerical property of the selected quantum system and operating point.

Therefore, QuTiP is used to determine the receiver response rather than to manufacture an analytical distortion equation.

27 Final Master Simulation Flow

The complete simulation is

$$\begin{aligned}
 \text{SNR grid point} &\rightarrow \gamma \rightarrow P_{\text{Rx}} \\
 &\rightarrow E_{\text{RF}} \\
 &\rightarrow \Omega_{\text{RF}} \\
 &\rightarrow \begin{cases} \text{Conventional AWGN,} \\ \text{LO-free QuTiP AT response,} \\ \text{LO-dressed QuTiP nonlinear response} \end{cases} \\
 &\rightarrow \text{receiver noise} \\
 &\rightarrow \text{symbol detector} \\
 &\rightarrow \text{SER.}
 \end{aligned} \tag{96}$$

Thus,

$$\boxed{\text{Analytical communication model} + \text{numerical quantum-response model} + \text{Monte-Carlo detection} = \text{complete}} \quad (97)$$

28 Minimal Execution Commands

```

1 # 1. Conventional validation
2 python validation/validate_conventional.py
3
4 # 2. Four-level QuTiP test
5 python quantum/four_level.py
6
7 # 3. LO-free EIT/AT response
8 python quantum/lo_free.py
9
10 # 4. LO-free lookup table
11 python quantum/response_tables.py --mode lo-free
12
13 # 5. LO-free Monte Carlo
14 python communication/lo_free_mc.py
15
16 # 6. LO-dressed nonlinear response
17 python quantum/lo_dressed.py
18
19 # 7. LO-dressed lookup table
20 python quantum/response_tables.py --mode lo-dressed
21
22 # 8. LO-dressed Monte Carlo
23 python communication/lo_dressed_mc.py
24
25 # 9. Final Fig. 9
26 python main.py

```

29 Final Objective

The final objective is to demonstrate three distinct physical behaviors.

$$\boxed{\text{Conventional: SNR} \uparrow \Rightarrow \text{SER} \downarrow} \quad (98)$$

$$\boxed{\text{LO-free: weak RF} \Rightarrow \text{unresolved AT splitting} \Rightarrow \text{distortion}} \quad (99)$$

$$\boxed{\text{LO-dressed: strong RF} \Rightarrow \text{nonlinear atomic response} \Rightarrow \text{distortion}} \quad (100)$$

The final architecture is therefore

$$\boxed{\begin{array}{l} \text{Conventional: analytical} + \text{Monte Carlo validation} \\ \text{LO-free: analytical} + \text{QuTiP AT characterization} + \text{Monte Carlo} \\ \text{LO-dressed: analytical} + \text{QuTiP nonlinear characterization} + \text{Monte Carlo} \end{array}} \quad (101)$$

The key physical mapping for both Rydberg receivers is

$$\boxed{\text{SNR}_{\text{dB}} \rightarrow \gamma \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{QuTiP response.}} \quad (102)$$

No arbitrary K_{Ω} is introduced.